

Strategy Reference Card

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Four archetypes, four BRAIN expressions. QM01 Strategy Concepts.

Cross-Sectional Rank

At each date, rank all stocks. Long the top decile, short the bottom decile. Trade the spread.

Edge source Risk premium + behavioral
When it fails Factor crowding, regime shift
Capacity High (multi-billion)

BRAIN `rank(ts_returns(close, 20))`

Time-Series Momentum

For each stock, if 6 to 12 month return is positive, go long. If negative, go short. Decide on its own history.

Edge source Behavioral (underreaction)
When it fails Sharp reversals, choppy tape
Capacity High

BRAIN `ts_returns(close, 60) - ts_returns(close, 5)`

Mean Reversion

If a stock moved sharply today, bet on partial reversal tomorrow. Microstructure pays liquidity providers.

Edge source Microstructure (liquidity premium)
When it fails Strong trending markets
Capacity Low to medium

BRAIN `-ts_zscore(returns, 5)`

Statistical Arbitrage

Strip out the common factor between similar assets. Trade the residual when it stretches beyond historical norm.

Edge source Information + microstructure
When it fails Cointegration breaks (M&A, sector split)
Capacity Medium

BRAIN `group_neutralize(-ts_zscore(returns, 20), subindustry)`